

Mental Health & Blood Pressure Correlation Analysis

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Research Question and Data Source

Hypertension is a major risk factor for cardiovascular disease, and blood pressure is influenced by both biological and psychosocial factors. Poor mental health, including stress and depressive symptoms, may increase blood pressure through behavioral and physiological pathways. Understanding the relationship between mental health and blood pressure can help identify early prevention targets.

*# **Research question***

Among adults aged 18 years and older in the NHANES teaching dataset, is the number of mentally unhealthy days in the past 30 days associated with systolic blood pressure, after adjusting for age, gender, race, BMI, and poverty status?

Data Source and Study Population

Data were obtained from the `NHANES` R package, which contains a curated sample of approximately 10,000 NHANES participants for educational use. For this project, we treat the data as if they come from a simple random sample.

We restrict the sample to adults (Age ≥ 18) with non-missing data on systolic blood pressure, mental health days, and our covariates.

Clean Data

```
nhanes_adult <- NHANES %>%  
  filter(Age >= 18) %>%  
  select(  
    Age, Gender, Race1, Race3, Poverty,  
    BMI, BPSysAve, DaysMentHlthBad  
  ) %>%  
  mutate(  
    # Prefer Race3 when available, otherwise Race1  
    Race = ifelse(is.na(Race3), as.character(Race1), as.character(Race3)),  
    Race = factor(Race)  
  ) %>%  
  filter(  
    !is.na(BPSysAve),
```

```

!is.na(DaysMentHlthBad),
!is.na(Age),
!is.na(Gender),
!is.na(Race),
!is.na(BMI),
!is.na(Poverty)
)

nrow(nhanes_adult)

## [1] 6097

summary(nhanes_adult)

##      Age      Gender      Race1      Race3      Poverty
##  Min.   :18.00  female:3000  Black   : 630  Asian   : 157  Min.
## 1st Qu.:31.00  male  :3097  Hispanic: 323  Black   : 311  1st
##  Median :46.00                Mexican : 472  Hispanic: 186  Median
##  Mean    :46.49                White   :4241  Mexican : 210  Mean
## 3rd Qu.:60.00                Other   : 431  White   :2125  3rd
##  Max.    :80.00                Other   :   84  Max.
##                                     NA's   :3024
##      BMI      BPSysAve  DaysMentHlthBad      Race
##  Min.   :15.02  Min.    : 78.0  Min.    : 0.000  Black   : 630
## 1st Qu.:24.07  1st Qu.:110.0  1st Qu.: 0.000  Hispanic: 323
##  Median :27.77  Median :119.0  Median : 0.000  Mexican : 472
##  Mean    :28.80  Mean    :120.8  Mean    : 4.111  Other   : 431
## 3rd Qu.:32.22  3rd Qu.:129.0  3rd Qu.: 4.000  White   :4241
##  Max.    :81.25  Max.    :226.0  Max.    :30.000
##

```

Hypothesis

```

## **Dependent variable (outcome):**
# Systolic blood pressure (`BPSysAve`, mm Hg): Continuous variable
# representing the average systolic blood pressure (mm Hg) across up to three
# blood pressure measurements taken during the physical examination. This is
# the primary cardiovascular outcome of interest.

## **Main independent variable (exposure):**
# Number of mentally unhealthy days in the past month (`DaysMentHlthBad`, 0-
# 30): Self-reported number of days (0-30) in the past month when the
# participant's mental health was "not good." Higher values indicate poorer
# recent mental well-being.

```

```

## **Covariates (confounders / adjustment variables):**
# Age (`Age`, years): Age in years at the time of screening. Adults aged 80
years or older are coded as 80. Age is included because blood pressure
increases naturally with age and strongly influences cardiovascular risk.
# Gender (`Gender`): Self-reported sex of the participant (male or female).
Gender is included because men and women differ biologically in blood
pressure regulation, especially across adulthood.
# Race/ethnicity (`Race`): Self-identified race/ethnicity category. In our
model, "Asian" is the reference group, and other categories include Hispanic,
Mexican, White, Black, and Other. Racial differences in blood pressure are
well-documented due to biological, social, and environmental factors.
# Body mass index (`BMI`, kg/m2): Calculated as weight (kg) divided by height
(m2). BMI is included because excess body weight increases blood pressure
through metabolic and cardiovascular strain.
# Poverty-income ratio (`Poverty`): A ratio of total household income to
federal poverty guidelines. Lower values represent greater socioeconomic
disadvantage. Poverty is included because socioeconomic status influences
both mental health and cardiovascular risk through access to care, chronic
stress, and health behaviors.

## We fit a **multiple linear regression model**:

## **Null hypothesis (H0):**
# After controlling for age, gender, race, BMI, and poverty, the number of
mentally unhealthy days is *not* associated with systolic blood pressure
(\(\beta_1 = 0\)).

# **Alternative hypothesis (Ha):**
# After controlling for these covariates, mentally unhealthy days *are*
associated with systolic blood pressure (\(\beta_1 \neq 0\)).

# We use a significance level of  $\alpha = 0.05$ .

```

Descriptive Statistics

```

nhanes_adult %>%
  summarise(
    n = n(),
    mean_age = mean(Age),
    sd_age = sd(Age),
    mean_bmi = mean(BMI),
    sd_bmi = sd(BMI),
    mean_bp = mean(BPSysAve),
    sd_bp = sd(BPSysAve),
    mean_ment = mean(DaysMentHlthBad),
    sd_ment = sd(DaysMentHlthBad)
  )

```

```
## # A tibble: 1 × 9
##       n mean_age sd_age mean_bmi sd_bmi mean_bp sd_bp mean_ment sd_ment
##   <int>   <dbl> <dbl>   <dbl> <dbl>   <dbl> <dbl>   <dbl>   <dbl>
## 1  6097    46.5  17.4    28.8   6.69    121.  16.7     4.11    7.89

# 95% CI for mean systolic BP:
bp_mean <- mean(nhanes_adult$BPSysAve)
bp_sd    <- sd(nhanes_adult$BPSysAve)
n        <- nrow(nhanes_adult)
se       <- bp_sd / sqrt(n)

lower_95 <- bp_mean - 1.96 * se
upper_95 <- bp_mean + 1.96 * se

c(mean = bp_mean,
  lower_95 = lower_95,
  upper_95 = upper_95)

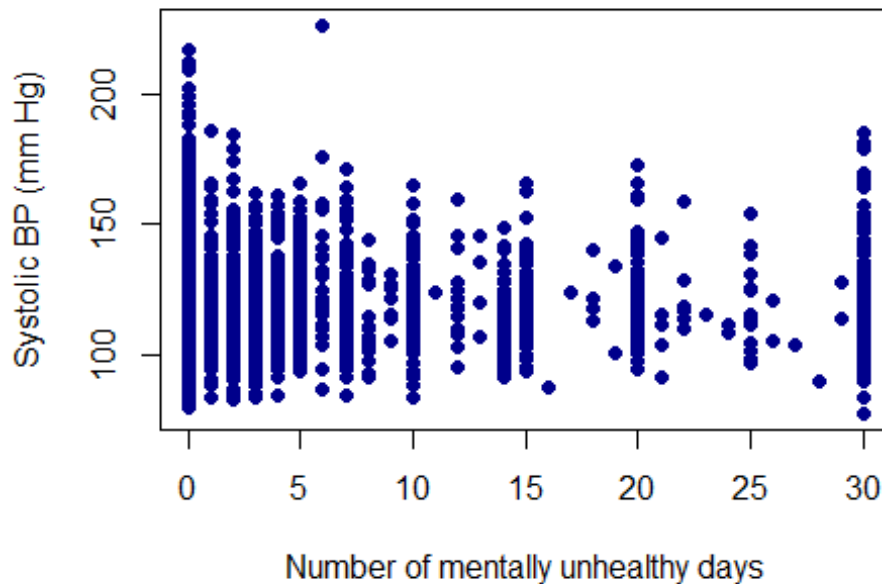
##       mean lower_95 upper_95
## 120.7518 120.3316 121.1721
```

Exploratory Plots

Scatterplot: Mental health days vs systolic BP

```
plot(nhanes_adult$DaysMentHlthBad, nhanes_adult$BPSysAve,
     main = "Systolic BP vs Mentally Unhealthy Days (past 30 days)",
     xlab = "Number of mentally unhealthy days",
     ylab = "Systolic BP (mm Hg)",
     pch = 16,
     col = "darkblue")
```

Systolic BP vs Mentally Unhealthy Days (past 30 da



The scatterplot of mentally unhealthy days and systolic blood pressure demonstrated substantial variability across all levels of mentally unhealthy days, with no obvious linear trend. Participants with 0 mentally unhealthy days showed a wide range of blood pressure values, which is expected given that this was the largest subgroup. Although individuals with very high levels of mentally unhealthy days (20–30 days) showed slightly higher blood pressure values, the overall relationship was weak, supporting the use of multiple regression to adjust for confounding factors.

Boxplot: : Systolic BP by mental health group

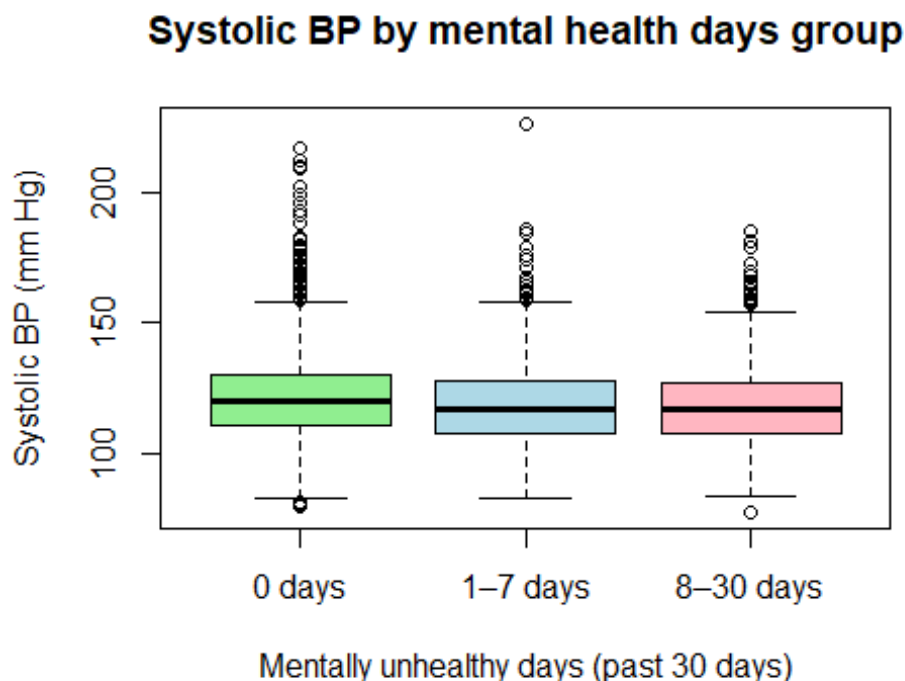
We can create a simple categorical version of mental health days:

*# 0 days
1–7 days
8–30 days*

```
nhanes_adult <- nhanes_adult %>%  
  mutate(  
    MentGroup = case_when(  
      DaysMentHlthBad == 0 ~ "0 days",  
      DaysMentHlthBad >= 1 & DaysMentHlthBad <= 7 ~ "1–7 days",  
      DaysMentHlthBad >= 8 ~ "8–30 days"  
    )  
  )  
  
table(nhanes_adult$MentGroup)
```

```
##
##    0 days   1-7 days 8-30 days
##    3508     1602    987

boxplot(BPSysAve ~ MentGroup,
        data = nhanes_adult,
        main = "Systolic BP by mental health days group",
        xlab = "Mentally unhealthy days (past 30 days)",
        ylab = "Systolic BP (mm Hg)",
        col = c("lightgreen", "lightblue", "lightpink"))
```



The distribution of systolic blood pressure appeared broadly similar across the three mental-health groups (0 days, 1-7 days, and 8-30 days of mentally unhealthy days). Median systolic BP increased slightly from the 0-day group to the 8-30-day group; however, there was substantial overlap in all three boxplots. Each group showed a wide range of blood pressure values, including extreme values (outliers), reflecting natural variability within the population. The 0-day group showed the highest number of outliers, which is expected because it had the largest sample size. Overall, the boxplot suggests only a weak unadjusted relationship between mentally unhealthy days and systolic blood pressure.

Covariate Assessment and Correlation Analysis

```
## Correlation matrix for continuous predictors
```

```

# Select numeric predictors and outcome
num_vars <- nhanes_adult[, c("BPSysAve", "Age", "BMI", "Poverty",
"DaysMentHlthBad")]

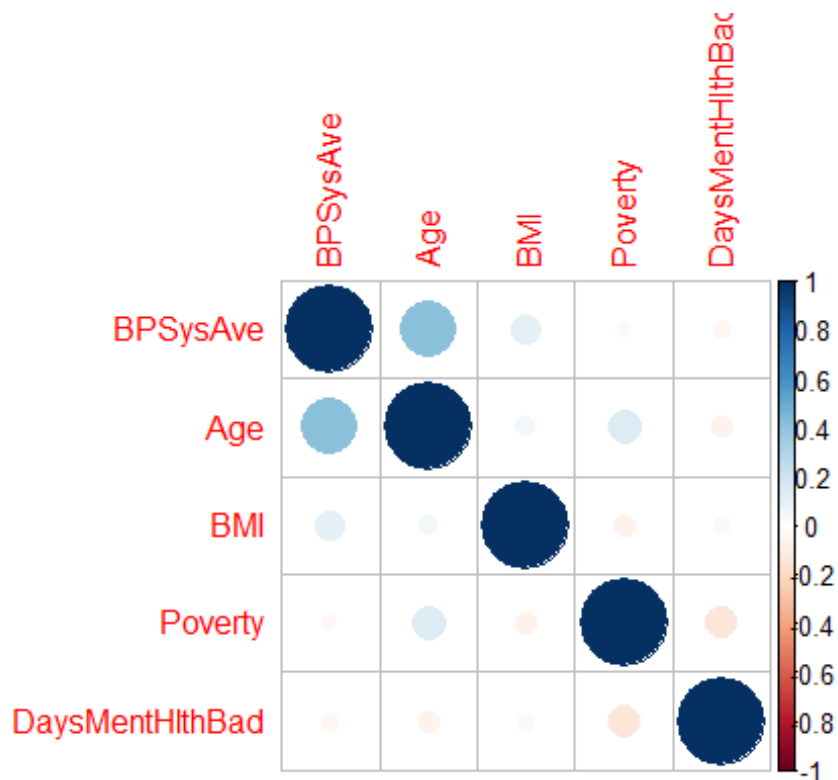
# Compute Pearson correlation matrix
cor_matrix <- cor(num_vars, use = "complete.obs")
library(corrplot)

## Warning: package 'corrplot' was built under R version 4.4.3

## corrplot 0.95 loaded

corrplot(cor_matrix)

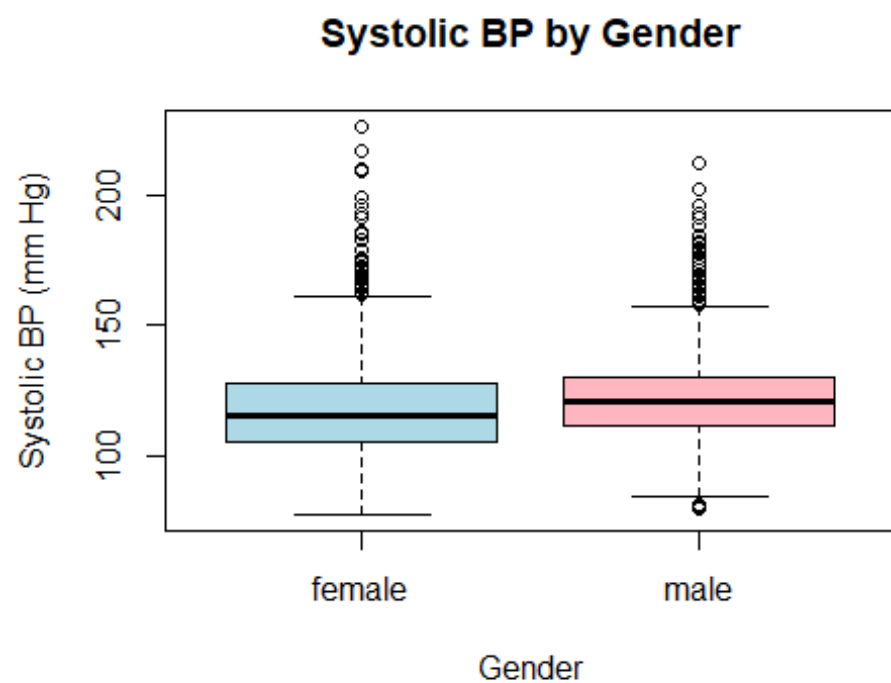
```



```

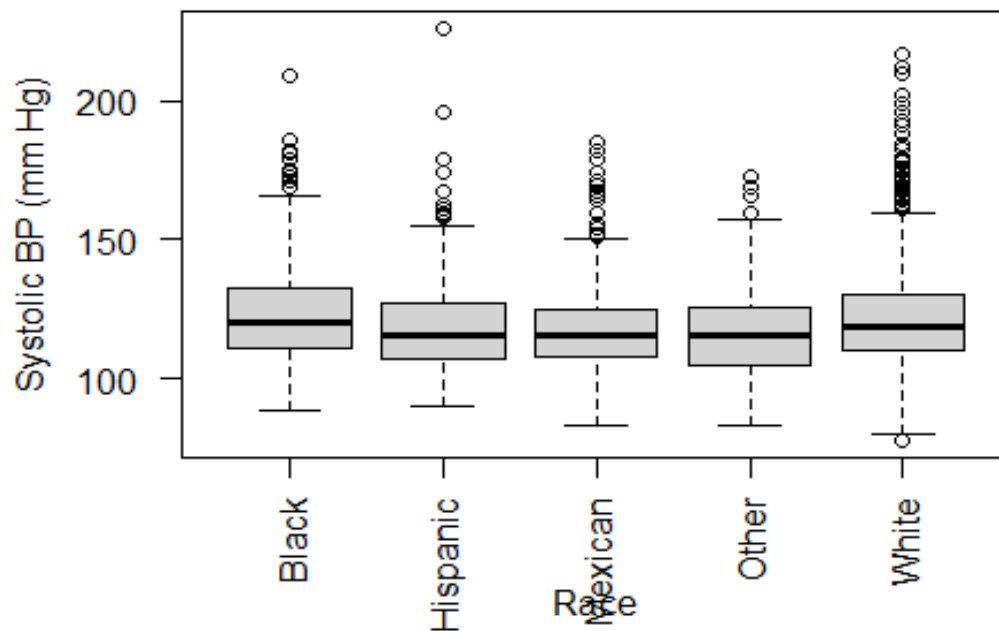
## Boxplots for categorical covariates
# Boxplot: BP by gender
boxplot(BPSysAve ~ Gender, data = nhanes_adult,
        main = "Systolic BP by Gender",
        ylab = "Systolic BP (mm Hg)",
        col = c("lightblue", "lightpink"))

```



```
# Boxplot: BP by race
boxplot(BPSysAve ~ Race, data = nhanes_adult,
        main = "Systolic BP by Race",
        ylab = "Systolic BP (mm Hg)",
        las = 2)
```

Systolic BP by Race



The correlation heat map demonstrated that systolic blood pressure was moderately positively correlated with both Age and BMI, consistent with known physiological patterns (older age and higher body mass are associated with higher blood pressure). Poverty showed a weak negative correlation with systolic BP, suggesting that individuals with higher poverty-income ratios tend to have slightly lower BP values, although the relationship was small. Days of mentally unhealthy days showed only a minimal correlation with systolic BP, indicating that its unadjusted association with blood pressure is weak.

Examination of categorical covariates through boxplots further supported the inclusion of gender and race in the regression model. The gender boxplot showed that males have noticeably higher systolic blood pressure compared to females, with a higher median and wider spread, suggesting a meaningful difference between groups.

Similarly, the race boxplot showed visible variation in systolic BP across racial categories, with some groups (e.g., Black participants) having higher median blood pressure and others (e.g., White or Mexican participants) showing lower central tendencies. Together, these findings indicate that Age, BMI, Poverty, Gender, and Race are important covariates to include in the multiple regression model, as they show measurable unadjusted relationships with systolic blood pressure. In contrast, days of mentally unhealthy days appear to have a weaker unadjusted relationship, underscoring the importance of evaluating its association with BP in an adjusted regression framework.

Multiple Linear Regression

We now fit one multiple linear regression model with systolic BP as the outcome and mentally unhealthy days as the main predictor, adjusting for age, gender, race, BMI, and poverty.

```
LM_final <- lm(BPSysAve ~ DaysMentHlthBad + Age + Gender +  
               Race + BMI + Poverty,  
               data = nhanes_adult)
```

```
summary(LM_final)
```

```
##  
## Call:  
## lm(formula = BPSysAve ~ DaysMentHlthBad + Age + Gender + Race +  
##     BMI + Poverty, data = nhanes_adult)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -55.986  -9.129   -0.826    7.964  101.357   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)   98.91868    1.22120   81.001  < 2e-16 ***  
## DaysMentHlthBad -0.04240    0.02465   -1.720  0.085493 .  
## Age           0.40903    0.01137   35.978  < 2e-16 ***  
## Gendermale     4.55061    0.38493   11.822  < 2e-16 ***  
## RaceHispanic  -2.13664    1.02361   -2.087  0.036897 *  
## RaceMexican   -3.04446    0.91555   -3.325  0.000889 ***  
## RaceOther     -4.02673    0.94316   -4.269  1.99e-05 ***  
## RaceWhite     -2.49215    0.65225   -3.821  0.000134 ***  
## BMI           0.19992    0.02899    6.895  5.93e-12 ***  
## Poverty       -0.91067    0.12316   -7.394  1.61e-13 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 14.93 on 6087 degrees of freedom  
## Multiple R-squared:  0.2059, Adjusted R-squared:  0.2047   
## F-statistic: 175.4 on 9 and 6087 DF,  p-value: < 2.2e-16
```

The multiple linear regression model predicting systolic blood pressure (BPSysAve) included mentally unhealthy days, age, gender, race, BMI, and poverty ratio as predictors. The intercept of 98.92 represents the expected systolic BP for an individual with all predictors equal to zero; while mathematically necessary, this value is not meaningful in practice because such a combination of characteristics (e.g., age = 0, BMI = 0) is unrealistic. After adjustment for all covariates, mentally unhealthy days were not significantly associated with systolic BP ($\beta = -0.04$, $p = 0.085$). Age showed a strong positive association, with each additional year corresponding to a 0.41 mm Hg increase in systolic BP, holding other variables constant. Males had systolic BP values 4.55 mm Hg higher than

females on average. Several race groups (Hispanic, Mexican, Other, and White) had significantly lower systolic BP compared with the reference group (Asian). Higher BMI was associated with higher systolic BP ($\beta = 0.20$), while higher poverty-income ratio was associated with lower systolic BP ($\beta = -0.91$). Overall, the model explained about 20% of the variation in systolic BP (Adjusted $R^2 = 0.2047$), and the overall F-test indicated that the set of predictors jointly contributed significantly to the model ($p < 2.2e-16$).

Clean coefficient table

```
tidy(LM_final, conf.int = TRUE) |>
  mutate(across(where(is.numeric), ~ round(.x, 3)))
```

A tibble: 10 × 7

##	term	estimate	std.error	statistic	p.value	conf.low	conf.high
##	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
##	1 (Intercept)	98.9	1.22	81.0	0	96.5	101.
##	2 DaysMentHlthBad	-0.042	0.025	-1.72	0.085	-0.091	0.006
##	3 Age	0.409	0.011	36.0	0	0.387	0.431
##	4 Gendermale	4.55	0.385	11.8	0	3.80	5.30
##	5 RaceHispanic	-2.14	1.02	-2.09	0.037	-4.14	-0.13
##	6 RaceMexican	-3.04	0.916	-3.32	0.001	-4.84	-1.25
##	7 RaceOther	-4.03	0.943	-4.27	0	-5.88	-2.18
##	8 RaceWhite	-2.49	0.652	-3.82	0	-3.77	-1.21
##	9 BMI	0.2	0.029	6.90	0	0.143	0.257
##	10 Poverty	-0.911	0.123	-7.39	0	-1.15	-0.669

Checking for model assumptions

We can use the standard diagnostic plots for linear regression:

4-panel layout

```
par(mfrow = c(2, 2),
    mar = c(4.5, 4.5, 3, 1))
```

1) Residuals vs Fitted Plot

```
plot(LM_final, which = 1,
     pch = 16, col = "darkgray",
     main = "Residuals vs Fitted")
abline(h = 0, col = "red", lty = 2)
```

The residuals-versus-fitted plot shows a roughly horizontal band of points centered around zero, with no strong curvature or pattern. This suggests that the assumptions of linearity and mean-independent residuals are reasonably met. Some slight spreading at higher fitted values is visible, which is common in large epidemiological datasets, but not severe enough to indicate major heteroscedasticity.

2) Normal Q-Q Plot

```
plot(LM_final, which = 2,
     pch = 16, col = "darkgray",
```

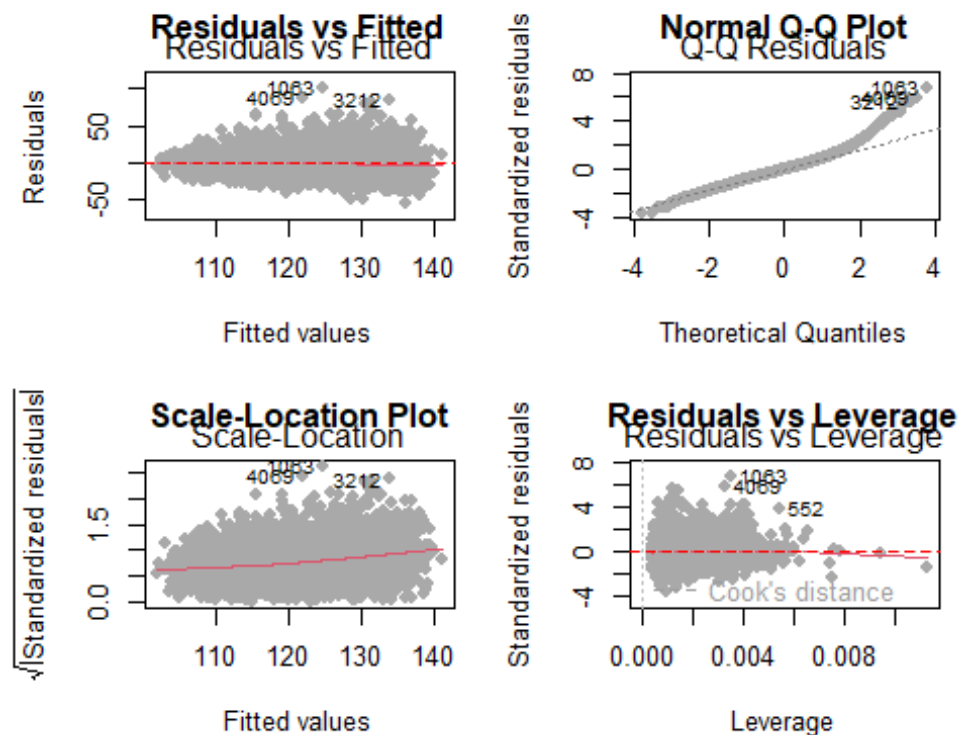
```

    main = "Normal Q-Q Plot")
## The Q-Q plot shows moderate deviation from the theoretical normal line at
both tails, indicating that the residuals are not perfectly normally
distributed. However, the central portion of the distribution aligns well
with the reference line. Given the large sample size ( $n > 6000$ ), mild tail
deviations are not concerning and do not meaningfully affect the validity of
the regression estimates.

# 3) Scale-Location Plot
plot(LM_final, which = 3,
     pch = 16, col = "darkgray",
     main = "Scale-Location Plot")
## The scale-location plot indicates mostly constant variance across fitted
values, with a slight upward trend at higher fitted values. This suggests
mild heteroscedasticity, but not severe enough to invalidate the model. The
pattern remains relatively tight and stable, supporting the assumption of
homoscedastic residuals.

# 4) Residuals vs Leverage Plot
plot(LM_final, which = 5,
     pch = 16, col = "darkgray",
     main = "Residuals vs Leverage")
abline(h = 0, col = "red", lty = 2)

```



The Leverage plot shows that most observations have Low Leverage, and there are no points with both high Leverage and large residuals. Cook's

distance values remain well below the threshold, indicating that no single observation exerts undue influence on the model. A few cases appear somewhat high-leverage but still fall within acceptable limits.

Examination of regression diagnostics indicated that the assumptions of multiple linear regression were reasonably met. The residuals demonstrated approximate linearity and constant variance, with no strong patterns evident. The Q-Q plot showed mild non-normality in the tails, expected with the large sample size, but the central distribution aligned well with the theoretical normal line. The scale-location plot suggested only slight heteroscedasticity. The leverage plot identified no influential observations. Overall, the diagnostic plots support the appropriateness of the multiple linear regression model for these data.

```
# Reset Layout  
par(mfrow = c(1, 1))
```

Conclusion

The goal of this study was to examine whether the number of mentally unhealthy days in the past 30 days was associated with systolic blood pressure among U.S. adults, after adjusting for demographic and health-related factors. Our primary hypothesis (H_1) stated that greater mentally unhealthy days would be associated with higher systolic blood pressure, while the null hypothesis (H_0) stated that there is no association.

After fitting a multiple linear regression model that controlled for age, gender, race, BMI, and poverty status, the coefficient for mentally unhealthy days was small and not statistically significant ($\beta = -0.042$, $p = 0.085$). Because this p -value is greater than the conventional 0.05 threshold, we fail to reject the null hypothesis. In practical terms, this means that there is no strong evidence that the frequency of mentally unhealthy days is directly associated with systolic blood pressure once other factors are accounted for.

Although mental health was not a significant predictor, several covariates were strongly associated with systolic blood pressure as expected. Age and BMI showed positive associations, indicating that older adults and individuals with higher BMI tended to have higher BP. Males had higher systolic BP compared with females, and several racial groups (Hispanic, Mexican, Other, White) had significantly lower BP compared with the reference group (Asian). Additionally, higher poverty-income ratio was linked to lower blood pressure, suggesting socioeconomic gradients in health.

The model explained approximately 20% of the variability in systolic BP (Adjusted $R^2 = 0.2047$), meaning that while the included predictors contribute meaningfully to BP differences, other unmeasured biological, behavioral, and environmental factors also play a substantial role.

Overall, our findings suggest that mental health alone does not significantly predict systolic blood pressure after adjusting for known physiological and demographic risk factors. This aligns with existing literature showing that mental health contributes more strongly to chronic stress pathways over long periods, while a single 30-day measure may not capture the complexity of these relationships.